

BioLayers AI: Microscopy AI Layer, Sprint Report

AI Model Developer sprint, Sep 8 – Oct 1, 2026 · Status as of Sep 28, 2026 · Research and educational system, not a clinical diagnostic

Summary. All six phases of the brief now have a working, benchmarked implementation. Segmentation (Phase 1), tracking (Phase 2) and phenotype measurement (Phase 3) are validated on public benchmark data with ground truth. Phenotype classification (Phase 4), biological linking (Phase 5) and confidence reporting (Phase 6) are built for the flagship case study: **glioblastoma cell migration → CXCL12/CXCR4 axis, with cancer-associated fibroblasts (CAFs) as the CXCL12 source**. No team imaging data (A549 or CAF) was available, so all results use public datasets. What remains is the website validation report, a one-command demo, the overlay handoff to the Software team, and the demo recording.

1. Status against the sprint plan

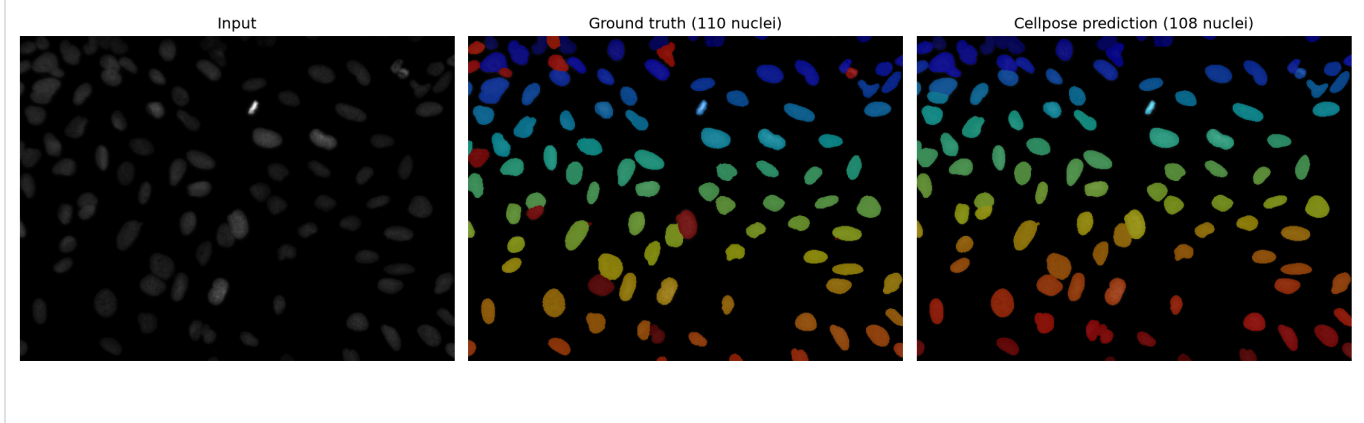
Sprint step (dates)	Status	What was delivered
1. Foundation (Sep 8–11)	Done	Cellpose 4.2.1.1 (pretrained cpsam_v2) on GPU; benchmark datasets BBBC039 + Cell Tracking Challenge PhC-C2DH-U373; evaluation code for IoU, Dice, precision, recall.
2. First working Cell Atlas (Sep 12–16)	Done	Segmentation on all 200 BBBC039 images (the full dataset; the plan asked for 250–350) with first metrics.
3. AI Microscopy (Sep 17–21)	Mostly done	Morphology extraction for 22,675 nuclei; metrics report with limitations. Overlay images exist, but a formal overlay handoff to the Software team is still open.
4. Time-lapse + Biological Linking (Sep 22–25)	Done	Tracking and migration on U373 time-lapse; Image → Cell → Phenotype → Biological Entity chain for the CAF/CXCL12 case study with evidence levels. (Finished Sep 28, 3 days late.)
5. Validation (Sep 26–28)	In progress	Per-phase validation reports exist (listed in section 9). The compact website version is not written yet.
6. Final polish (Sep 29–30)	Not started	One-command demo, stability check, 90-second recording.

2. Phase 1: Cell segmentation

Approach. Pretrained Cellpose model, used zero-shot (no fine-tuning), per the brief’s direction to use proven models rather than train from scratch.

Benchmark. BBBC039: 200 fluorescence images of U2OS nuclei (Hoechst stain) with hand-annotated masks, CC0 license. Detection is scored by matching each predicted nucleus to a real one (Hungarian matching, $\text{IoU} \geq 0.5$).

Metric (mean over 200 images)	Value	Metric	Value
Pixel IoU	0.942 ± 0.017	Object precision	0.980
Pixel Dice	0.970 ± 0.009	Object recall	0.966
Nuclei: correct / false / missed	22,212 / 463 / 766	Object F1	0.972
Inference time (RTX 5060)	723 ms / image	Mean IoU of matched nuclei	0.928



BBBC039 example: input image, hand-annotated ground truth (110 nuclei), Cellpose prediction (108 nuclei).

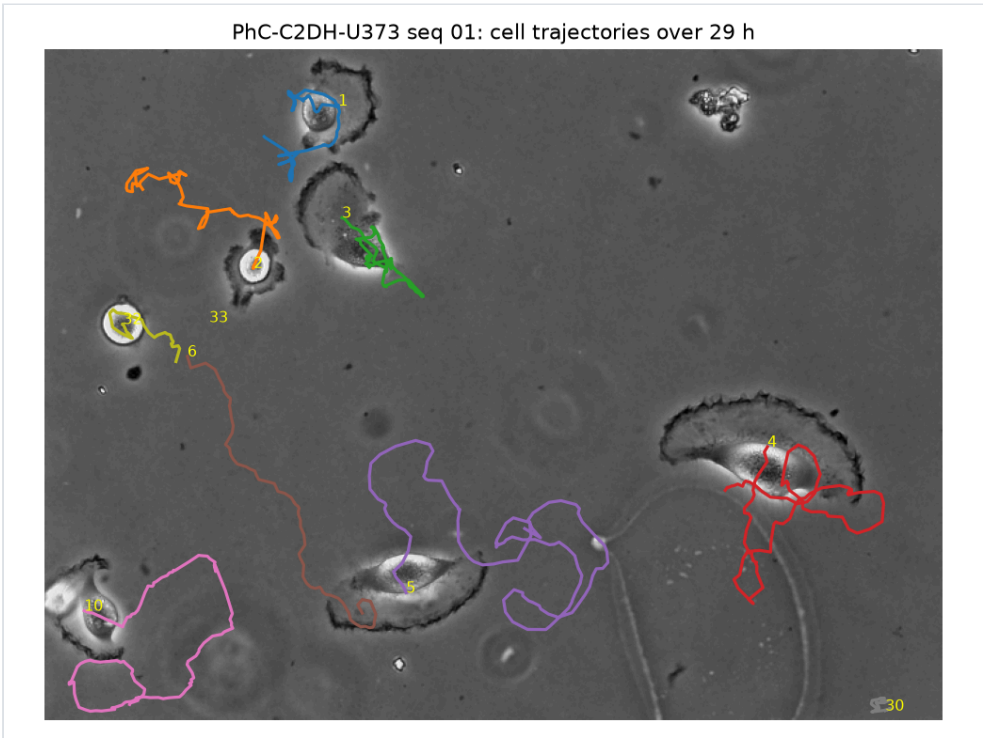
Limits. This measures nuclei in a DNA stain only. Whole-cell phase-contrast segmentation is scored separately in Phase 2 (CTC SEG 0.94 and 0.87).

3. Phase 2: Cell tracking and migration

Data. Cell Tracking Challenge PhC-C2DH-U373: phase-contrast time-lapse of glioblastoma-astrocytoma cells, 2 sequences × 115 frames (15 min/frame, 28.5 h each), with ground-truth tracks.

Method. Cellpose segments every frame. Cells are linked frame to frame by mask overlap (Hungarian matching), then brief segmentation flicker and short gaps are cleaned up. Each cell's migration distance, speed and directionality (net displacement ÷ path length) are measured.

Seq	Detection precision	Detection recall	Link accuracy	ID switches	CTC SEG	Divisions (pred / real)	Mean speed $\mu\text{m}/\text{min}$ (pred / real)
01	0.928	0.999	0.999	1	0.940	1 / 0	0.183 / 0.205
02	0.857	0.987	0.995	3	0.867	1 / 0	0.137 / 0.158



Sequence 01: tracked cell trajectories over 28.5 h.

Limitations.

- **Metric source.** We implemented the metrics following the CTC definitions. They are not scores from the official CTC evaluation software, so they are not comparable to the CTC leaderboard.
- **Division detection is not validated.** These sequences contain no real divisions. The detector reports 1 false division per sequence, and whether it finds real divisions is unmeasured. That needs a dataset with divisions, such as CTC Fluo-N2DL-HeLa.
- **Speeds read low.** Measured speeds are about 12% below ground truth. Speed is path length over time, so it is sensitive to position jitter, and ground-truth positions are hand-placed markers while ours are mask centroids.
- **Small sample.** Each sequence has only 6–12 cells, so one tracking error visibly moves the metrics.

4. Phase 3: Phenotype extraction

Features were measured from the segmentation masks for **22,675 nuclei** (BBBC039) and for every tracked U373 cell in every frame. All of these are direct measurements from segmented pixels.

Feature (BBBC039 nuclei)	Mean $\pm$ SD	Feature	Mean $\pm$ SD
Area (px)	627 $\pm$ 256	Aspect ratio	1.71 $\pm$ 0.61
Perimeter (px)	95.0 $\pm$ 23.6	Solidity	0.958 $\pm$ 0.025
Circularity	0.830 $\pm$ 0.092	Mean intensity	648 $\pm$ 217
Eccentricity	0.746 $\pm$ 0.138	Texture (GLCM)	contrast, homogeneity, energy, correlation

Migration distance, speed and division events come from Phase 2.

Not computed: nuclear/cytoplasmic ratio. It needs paired nucleus and cytoplasm channels, and neither dataset has both. It is reported as missing rather than approximated.

5. Phase 4: Phenotype classification

Approach. No labelled data exists for the five target classes, so classification is a set of **transparent threshold rules**, not a trained model. Each label records the rule that produced it. Its score is a **rule margin**: how far the deciding measurement is from its threshold. This is not a probability.

Class	Rule
Migrating	Net displacement $\geq$ 1 own cell diameter over $\geq$ 20 frames (5 h)
Quiescent	Net displacement $<$ 0.5 cell diameter over $\geq$ 20 frames, no division. An operational definition: true cell-cycle arrest is not visible in phase contrast.
Proliferating	Cell divides and both daughters persist $\geq$ 3 frames
Apoptotic/dead	Time-lapse: shrinks to $\leq$ 60% of its usual area and rounds up over its last 2 h. Single images: small, very bright nucleus (pyknotic-like).
Abnormal morphology	Area, circularity or solidity is a strong outlier in its population (robust $z \geq 3.5$)

Results, U373 time-lapse (22 tracked cells): 10 migrating, 2 quiescent, 1 apoptotic/dead, 1 proliferating, 4 abnormal morphology, 10 indeterminate (track too short, or moved between 0.5 and 1 diameter). A cell can carry more than one label.

Results, BBBC039 single images (22,675 nuclei): only shape-based classes can be judged from one image. 2,102 nuclei were flagged abnormal morphology and 413 apoptotic-like. Another 4,430 are cut by the image edge and were not classified.

How reliable are the labels? There are no expert labels to score them against. Two indirect checks were run, and one known error is listed:

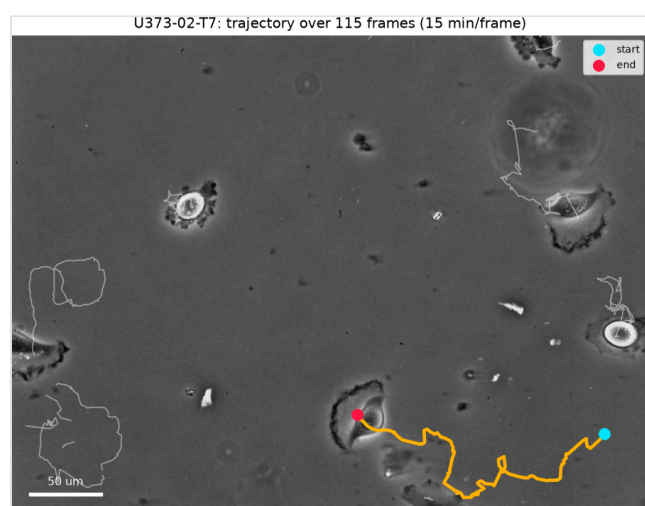
- **Tracking errors vs labels:** applying the same rule to the hand-annotated trajectories gives the same label for **10 of 14** matched cells. Of the 4 differences, 2 are one of our tracks following two different cells (an identity switch).
- **Flags vs segmentation errors:** 86% of abnormal-morphology flags are a single real nucleus in the BBBC039 annotations. However, 6.8% are two touching nuclei merged by segmentation, against 0.5% of unflagged nuclei, so part of the flag reflects segmentation error rather than biology.
- **Known false positive:** the one "proliferating" label comes from the false division detection in Phase 2.

6. Phase 5: Biological linking (flagship case study)

Case study choice. The sprint plan names CAF/CXCL12. The brief's other example, A549 → EGFR, would need A549 imaging we don't have. The public U373 migration movies fit the CXCL12/CXCR4 literature on glioma cell migration, so the chain was built on them.

Three evidence levels, never merged:

- **Observed in image:** measured from segmented pixels (for example distance travelled).
- **Supported by literature:** published findings about the cell type or pathway, never about the specific imaged cell.
- **Inferred by model:** rule labels and hypotheses.



Flagship cell U373-02-T7 (orange): 115 frames, 28.5 h. Other tracked cells in grey.

Image → Cell OBSERVED IN IMAGE

Segmented in every frame; mean cell probability 0.93.

Cell → Measured phenotype OBSERVED IN IMAGE

Net displacement 176 μm (4.3 cell diameters), path 331 μm, speed 0.19 μm/min, directionality 0.53.

Phenotype label INFERRED BY MODEL

Migrating (rule: moved ≥ 1 cell diameter). Rule margin 0.99.

Biological entity: CXCL12/CXCR4 SUPPORTED BY LITERATURE

Glioma cells express CXCR4, and CXCL12 drives their chemotaxis and invasion (Zhou 2002; Ehtesham 2006; Rempel 2000).

Biological entity: CAFs SUPPORTED BY LITERATURE

CAFs secrete CXCL12, which acts through CXCR4 (Orimo 2005, breast carcinoma).

Hypothesis INFERRED BY MODEL

This cell's migration may involve CXCL12/CXCR4 signalling. Not tested in these images.

Literature base. A curated knowledge base of 5 papers. Every statement was checked against the paper's PubMed abstract.

Link	Source
CXCR4 on glioma cells; CXCL12 drives chemotaxis	Zhou et al., J Biol Chem 2002 (PMID 12388552)
CXCR4 mediates glioma invasion toward CXCL12	Ehtesham et al., Oncogene 2006 (PMID 16407848)
CXCL12/CXCR4 rise with glioma grade	Rempel et al., Clin Cancer Res 2000 (PMID 10656438)
CAFs secrete CXCL12 acting via CXCR4	Orimo et al., Cell 2005 (PMID 15882617). Shown in breast carcinoma, not glioma.
Caveat: cell line identity	Torsvik et al., Cancer Med 2014 (PMID 24810477). Many "U-373" stocks are actually U-251.

What this does not show. The movies contain no CXCL12, no CXCR4 staining and no fibroblasts, and one cell moving persistently in an unstimulated dish is not evidence of chemotaxis. The per-cell link is therefore stated only as a hypothesis. Testing it would need a CXCL12 gradient experiment and a CXCR4-blocking control.

7. Phase 6: AI confidence and provenance

Every AI-derived value in the output points to a model card with the six fields the brief requires. Benchmark numbers are read directly from the Phase 1–2 result files.

Component	Model / version	Confidence shown	Evaluated on / metric	Main limitation
Segmentation	Cellpose cpsam_v2, v4.2.1.1	Per cell: mean Cellpose cell probability	BBBC039 IoU 0.942, P 0.980, R 0.966; U373 SEG 0.94 / 0.87	Score is uncalibrated; only two datasets tested
Tracking	Overlap tracker, code 44ec666	Per track: mean mask overlap between frames	U373 link accuracy 0.999 / 0.995	Division recall unmeasured
Phenotype rules	Rules v0.1.0	Per label: rule margin	10/14 agreement with ground-truth trajectories	No expert labels
Literature links	Knowledge base v0.1.0	Citation per statement	Checked against PubMed abstracts	Only "migrating" has links so far

The per-cell output (`reports/phase5_cell_atlas_u373.json`) lists every tracked cell with its measurements, labels, literature links and hypotheses. Each item is tagged with its evidence level and model card.

8. Remaining work and open items

When	Work
Sep 29	Compact validation report for the website; one-command demo (image → cells → phenotype → linked biology); segmentation overlay package for the Software team.
Sep 30	Stability run of the demo; 90-second video recording.
Oct 1	Release.

Open items for the team:

- **Website:** who publishes the validation report on the website, and in what format (Markdown, PDF or web page)?

- **Software handoff:** which overlay format the Software team needs (PNG overlays, label masks, or per-cell JSON).
- **Team data:** if A549 or CAF co-culture imaging becomes available, the pipeline can run on it directly. The nuclear/cytoplasmic ratio also needs two-channel images.
- **Repository:** the code is currently in github.com/Maksatb07/biolayers-ai and can be moved to a shared team repository.

9. Known gaps after release

- Validate division detection on a dataset that contains divisions (e.g. CTC Fluo-N2DL-HeLa).
- Get expert-labelled cells to measure phenotype label accuracy; the rules are currently unvalidated.
- Test segmentation on further stains and cell types; only two public datasets are benchmarked.
- Add literature links for phenotypes beyond migration, and the A549 → EGFR example.

Per-phase reports in the repository: [phase1_validation_report.md](#), [phase2_tracking_report.md](#), [phase3_phenotype_report.md](#), [phase4_classification_report.md](#), [phase5_6_case_study_report.md](#) (all in [reports/](#)).